# Visual Validation Report
## Connectomics Split-Detection Pipeline

Source: output/xpress_training
Segmentation: data/xpress/baseline_seg_training.h5
Role: input baseline segmentation (automated, imperfect)
Fragment IDs are segment IDs from this volume.
Raw EM: not provided — Panel 1 shows neutral segmentation
Skeleton: data/xpress/XPRESS_training_skels.npz
Role: ground-truth axon traces used to define merge pairs.
A pair (A, B) is GT-positive iff a skeleton edge crosses
the boundary between segment A and segment B.

Candidates: 369,570 total | Accepted: 368,181 | Rejected: 1,389

Sample shown: 30 candidates

- Ground-truth TP: 5
- High-confidence accepted: 5
- Low-confidence accepted: 5
- Borderline rejected: 5
- GT FN: 5
- Random rejected: 5

Ground-truth: 1,499 skeleton-derived merge pairs loaded — TP/FP/FN/TN shown in Panel 3

### ─── Ground-truth semantics ───

'GT SHOULD MERGE' means the (label_a, label_b) pair appears in the skeleton-derived merge set. The set is built by mapping each skeleton edge endpoint to a voxel in the segmentation; if the two endpoints fall in different non-background segments, that (seg_a, seg_b) pair is added (edge-crossing criterion). The dashed line in Panel 3 is a GT-positive *indicator* connecting fragment centroids — it is NOT the skeleton edge itself. Verify the anatomical reality by checking the yellow skeleton overlay in Panels 1 and 3: the skeleton should pass through both fragments.

Pipeline decisions are based on skeleton geometry (alignment, continuity, curvature) and rule-based filters. Ground-truth labels are used only for post-hoc evaluation — they play no role in the pipeline decision.

### ─── Panel layout (3 columns per candidate row) ───

Panel 1 — Raw / Neutral:
Grayscale EM (if available) or neutral segmentation.
Yellow dots/lines = skeleton nodes/edges near the Z-slice.
Use this to assess: what does the tissue look like?

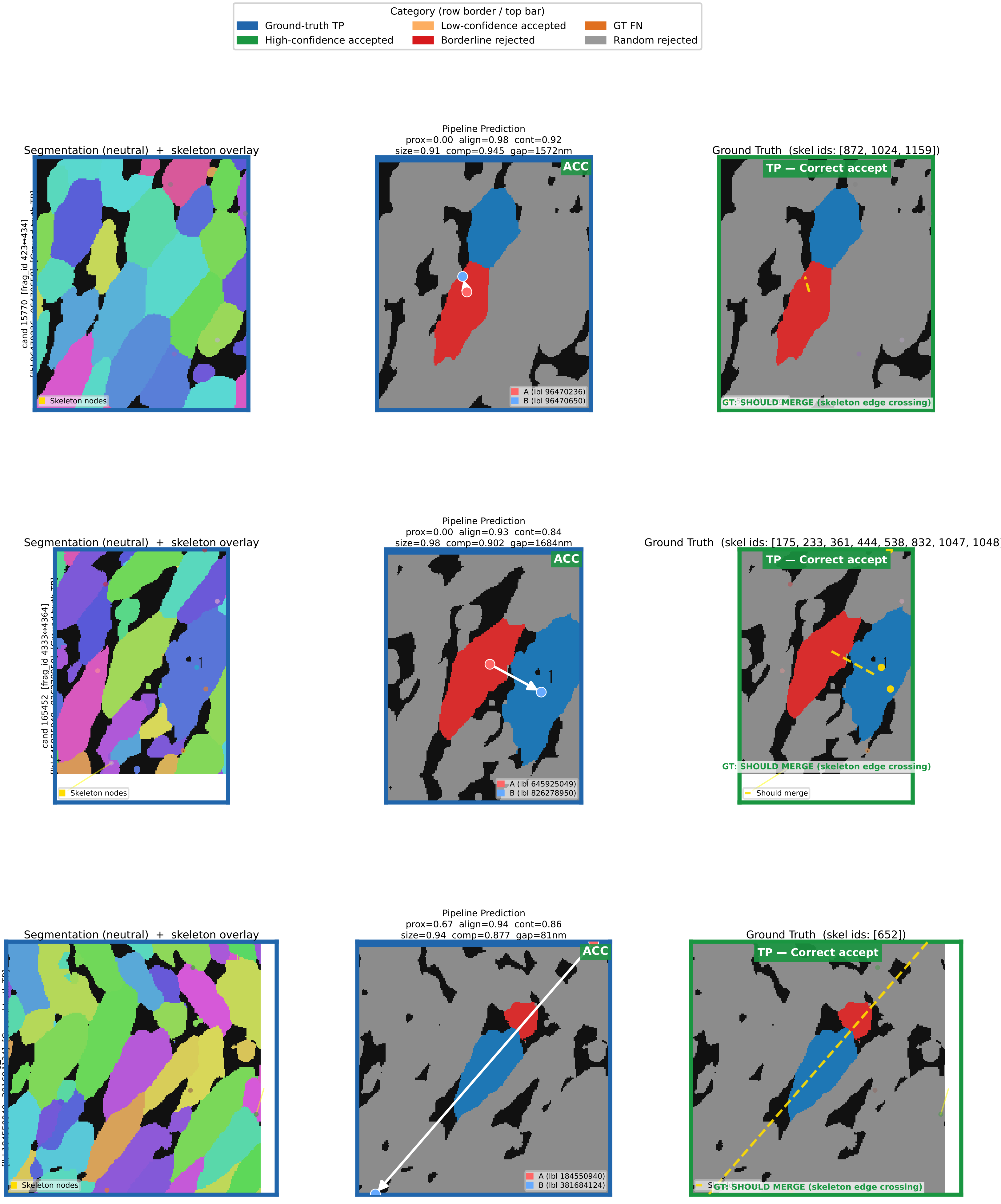
Panel 2 — Pipeline Prediction:
Red = Fragment A, Blue = Fragment B, Gray = others.
White arrow = proposed merge direction A→B.
Badge: ACC (green) or REJ (red).
Title shows all 5 score components + gap distance.

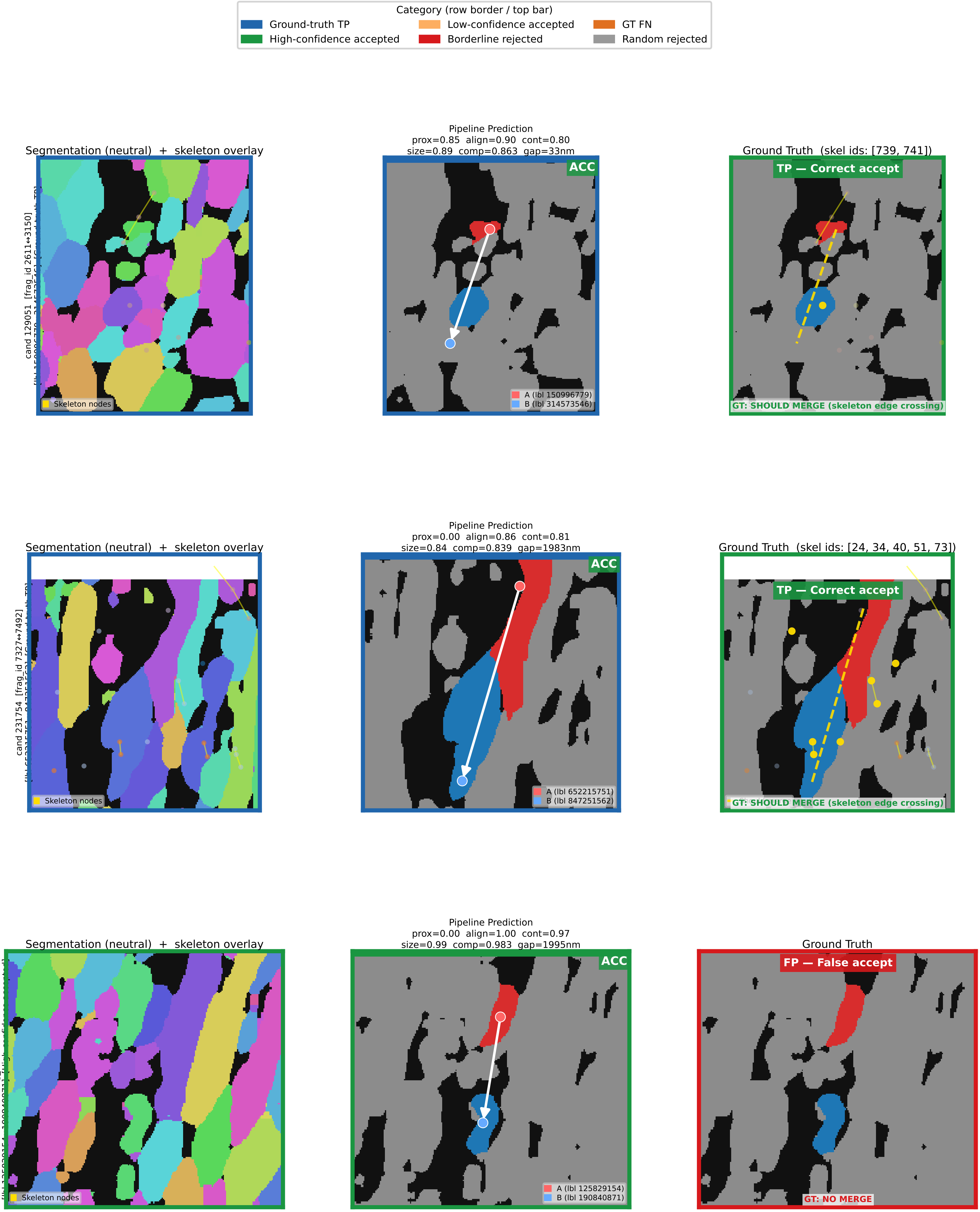
Panel 3 — Ground Truth:
Same seg coloring + ground-truth overlay.
TP (green) = correct accept — pipeline & GT agree: merge
FP (red) = false accept — pipeline merged, GT says no
FN (orange) = missed merge — pipeline rejected, GT says merge
TN (gray) = correct reject — pipeline & GT agree: no merge
Bright yellow skeleton = neuron(s) passing through fragment A or B.
Dashed line = GT-positive indicator (TP/FN only); verify with skeleton overlay, not the dashed line alone.

TP
FP
FN
TN

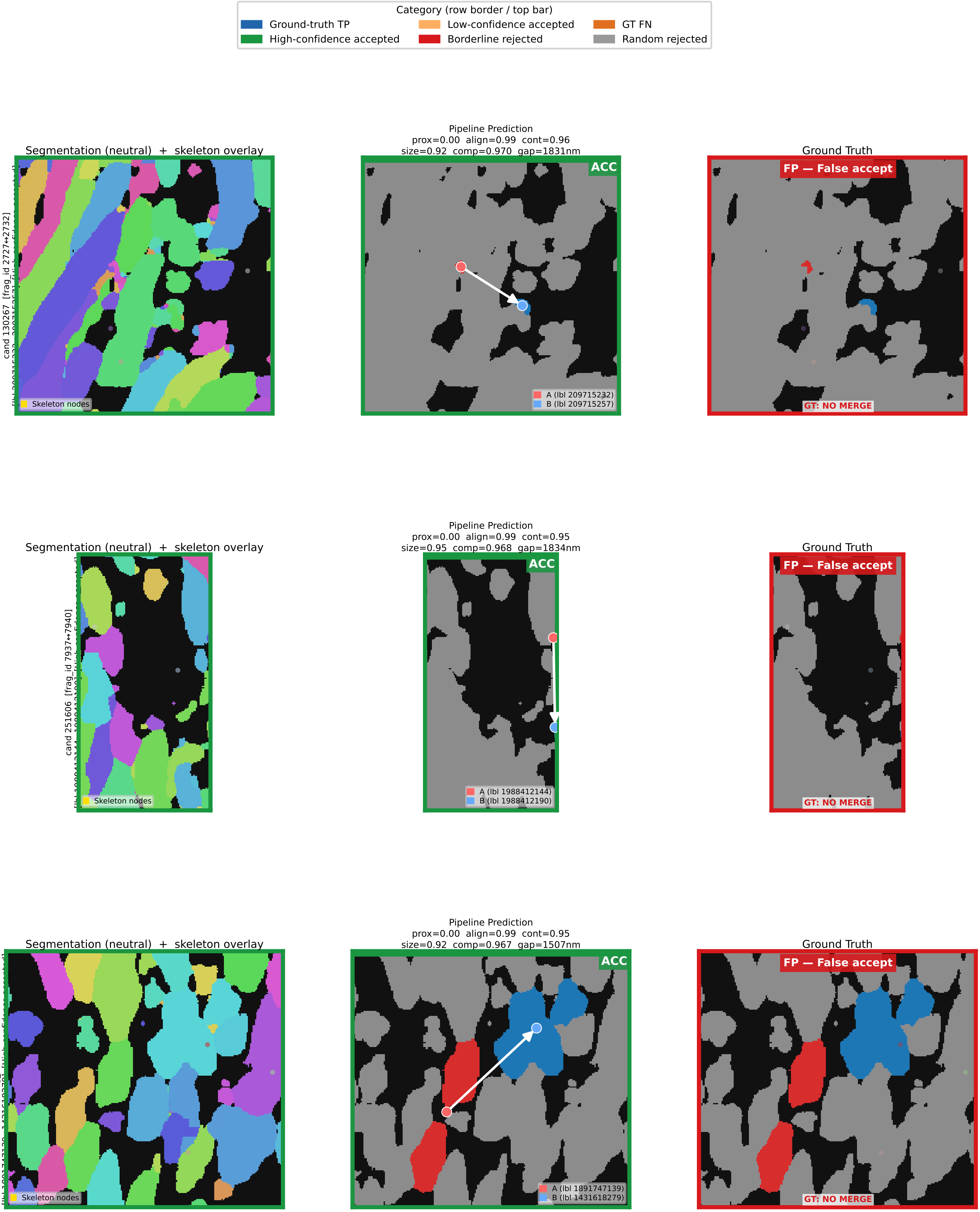
### ─── How to inspect ───

1. Look at Panel 1 to understand the tissue context.
2. In Panel 2, check if fragments A and B plausibly belong to the same neuron (do they touch? Are shapes continuous?).
3. In Panel 3, verify the ground truth: does the skeleton cross from one fragment into the other? For FP/FN cases, examine whether the pipeline score values explain the error.

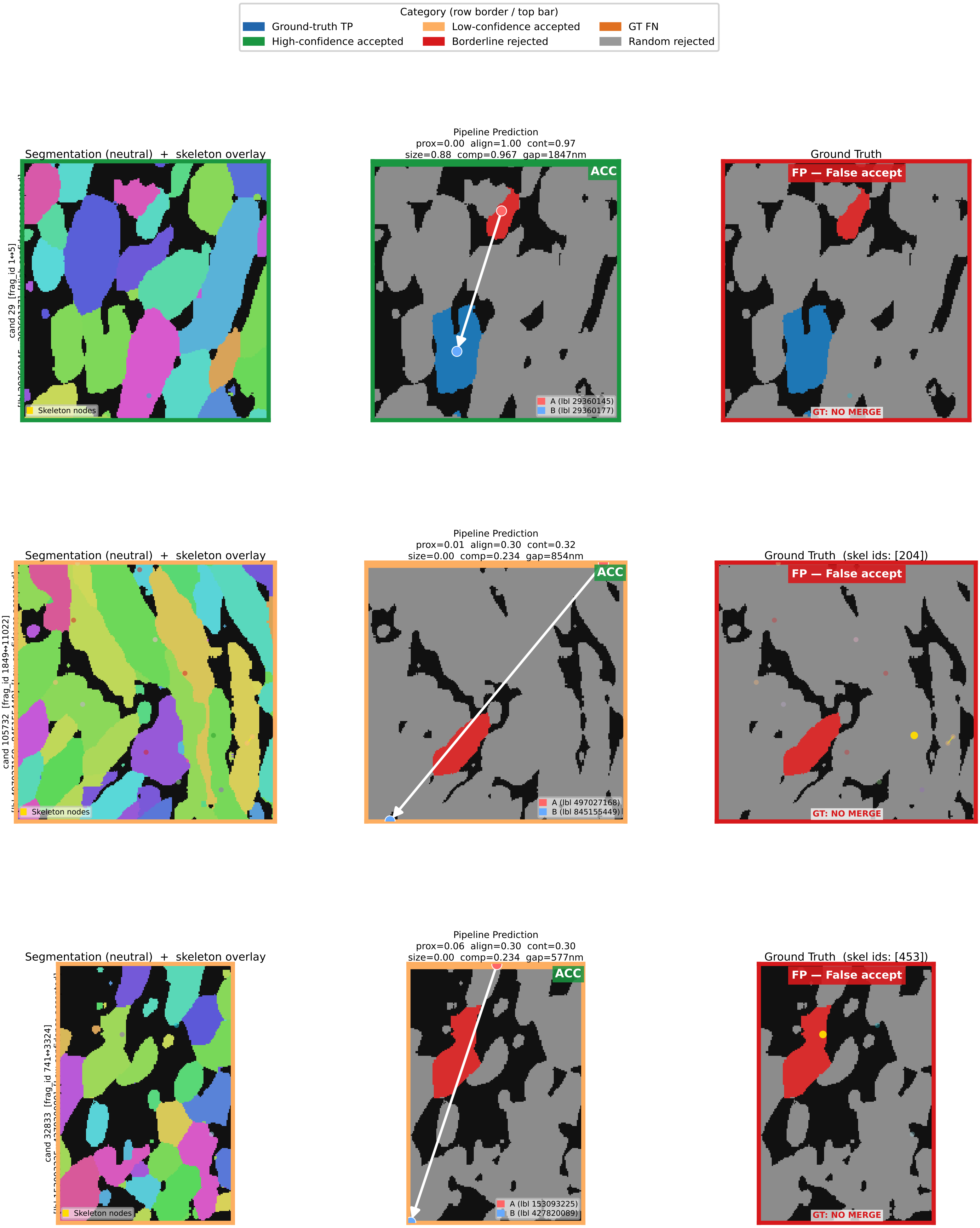

Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 15770 [frag_id 423↔434]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.98 cont=0.92
size=0.91 comp=0.945 gap=1572nm
ACC
A (lbl 96470236)
B (lbl 96470650)
Ground Truth (skel ids: [872, 1024, 1159])
TP — Correct accept
GT: SHOULD MERGE (skeleton edge crossing)
Segmentation (neutral) + skeleton overlay
cand 165452 [frag_id 4333↔4364]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.93 cont=0.84
size=0.98 comp=0.902 gap=1684nm
ACC
A (lbl 645925049)
B (lbl 826278950)
Ground Truth (skel ids: [175, 233, 361, 444, 538, 832, 1047, 1048])
TP — Correct accept
GT: SHOULD MERGE (skeleton edge crossing)
Should merge
Segmentation (neutral) + skeleton overlay
cand 73819 [frag_id 1228↔8303]
Skeleton nodes
Pipeline Prediction
prox=0.67 align=0.94 cont=0.86
size=0.94 comp=0.877 gap=81nm
ACC
A (lbl 184550940)
B (lbl 381684124)
Ground Truth (skel ids: [652])
TP — Correct accept
GT: SHOULD MERGE (skeleton edge crossing)



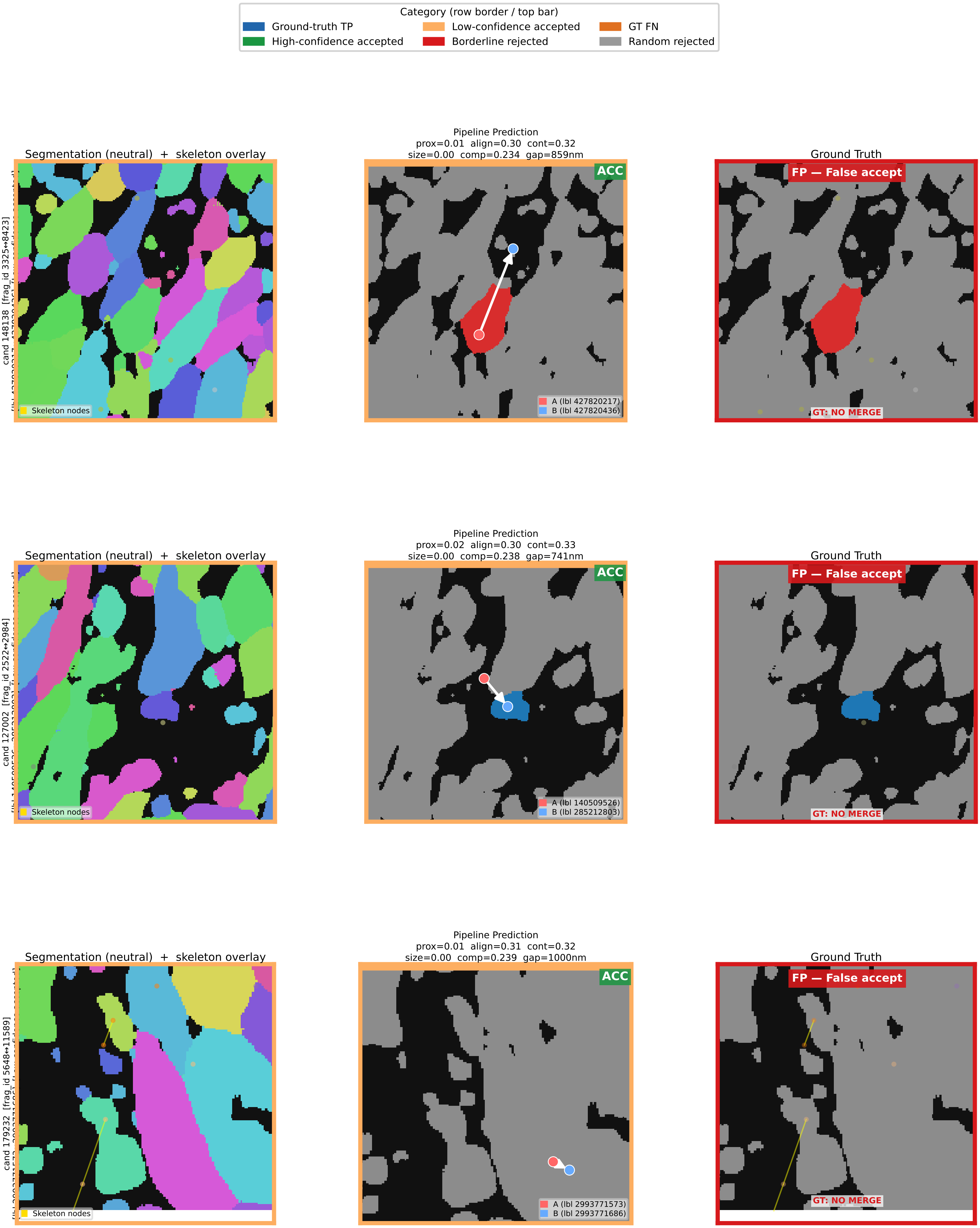
Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 129051 [frag_id 2611↔3150]
Skeleton nodes
Pipeline Prediction
prox=0.85 align=0.90 cont=0.80
size=0.89 comp=0.863 gap=33nm
ACC
A (lbl 150996779)
B (lbl 314573546)
Ground Truth (skel ids: [739, 741])
TP — Correct accept
GT: SHOULD MERGE (skeleton edge crossing)
Segmentation (neutral) + skeleton overlay
cand 231754 [frag_id 7327↔7492]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.86 cont=0.81
size=0.84 comp=0.839 gap=1983nm
ACC
A (lbl 652215751)
B (lbl 847251562)
Ground Truth (skel ids: [24, 34, 40, 51, 73])
TP — Correct accept
GT: SHOULD MERGE (skeleton edge crossing)
Segmentation (neutral) + skeleton overlay
cand 59830 [frag_id 1065↔1234]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=1.00 cont=0.97
size=0.99 comp=0.983 gap=1995nm
ACC
A (lbl 125829154)
B (lbl 190840871)
Ground Truth
FP — False accept
GT: NO MERGE

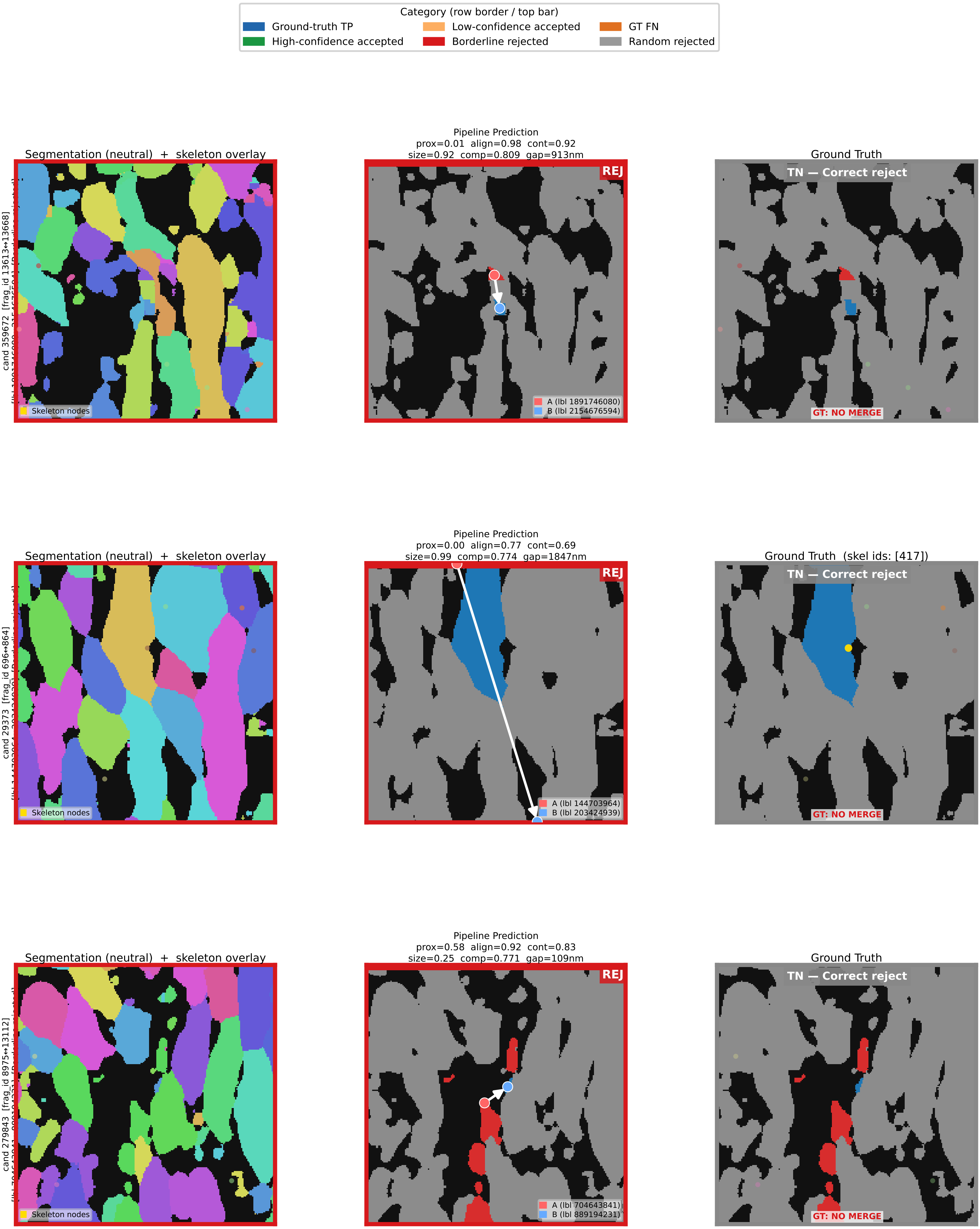


Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 130267 [frag_id 2727↔2732]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.99 cont=0.96
size=0.92 comp=0.970 gap=1831nm
ACC
A (lbl 209715232)
B (lbl 209715257)
Ground Truth
FP — False accept
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 251606 [frag_id 7937↔7940]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.99 cont=0.95
size=0.95 comp=0.968 gap=1834nm
ACC
A (lbl 1988412144)
B (lbl 1988412190)
Ground Truth
FP — False accept
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 333604 [frag_id 12109↔12368]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.99 cont=0.95
size=0.92 comp=0.967 gap=1507nm
ACC
A (lbl 1891747139)
B (lbl 1431618279)
Ground Truth
FP — False accept
GT: NO MERGE

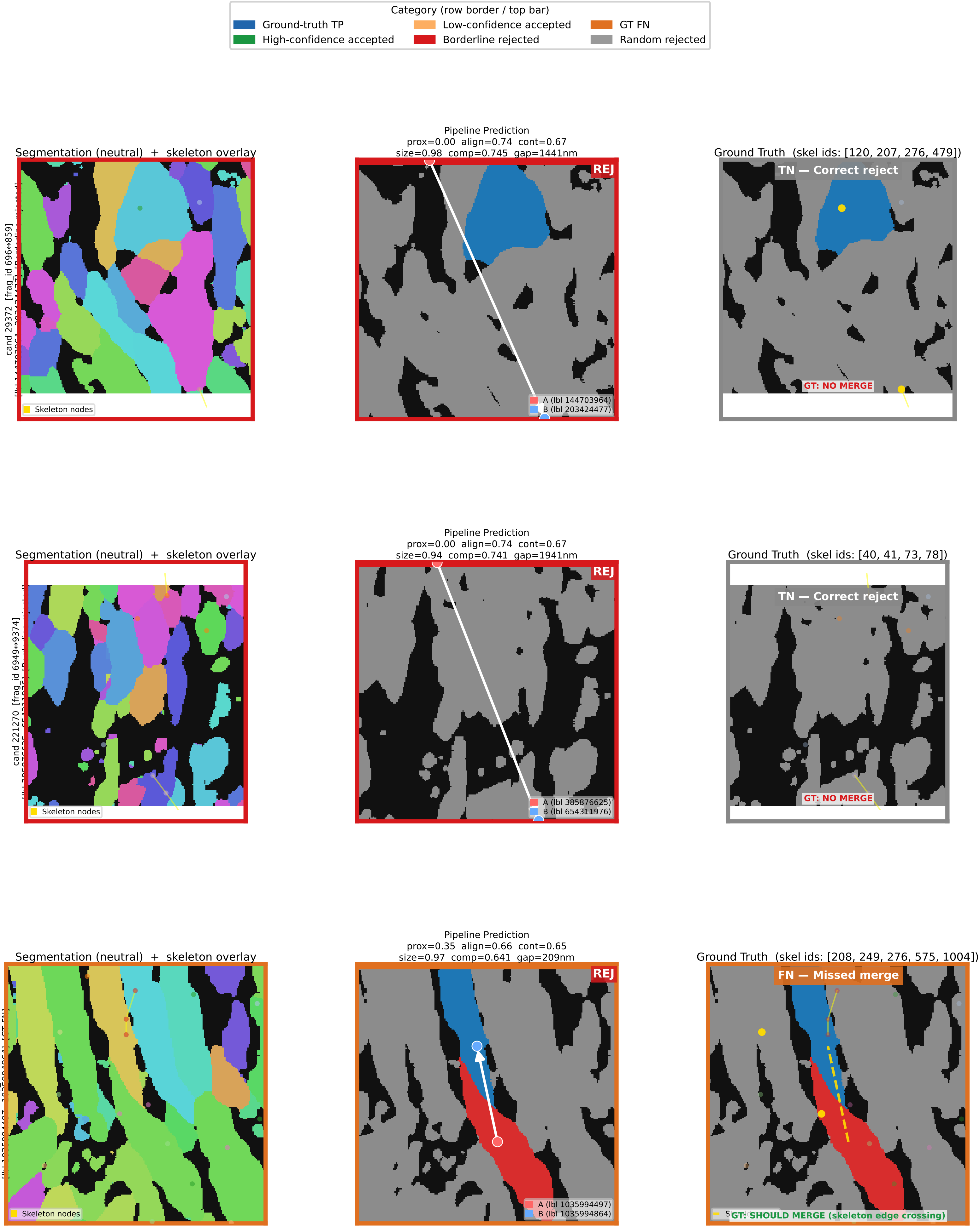


Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 29 [frag_id 1↔5]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=1.00 cont=0.97
size=0.88 comp=0.967 gap=1847nm
ACC
A (lbl 29360145)
B (lbl 29360177)
Ground Truth
FP — False accept
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 105732 [frag_id 1849↔11022]
Skeleton nodes
Pipeline Prediction
prox=0.01 align=0.30 cont=0.32
size=0.00 comp=0.234 gap=854nm
ACC
A (lbl 497027168)
B (lbl 845155449)
Ground Truth (skel ids: [204])
FP — False accept
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 32833 [frag_id 741↔3324]
Skeleton nodes
Pipeline Prediction
prox=0.06 align=0.30 cont=0.30
size=0.00 comp=0.234 gap=577nm
ACC
A (lbl 153093225)
B (lbl 427820089)
Ground Truth (skel ids: [453])
FP — False accept
GT: NO MERGE

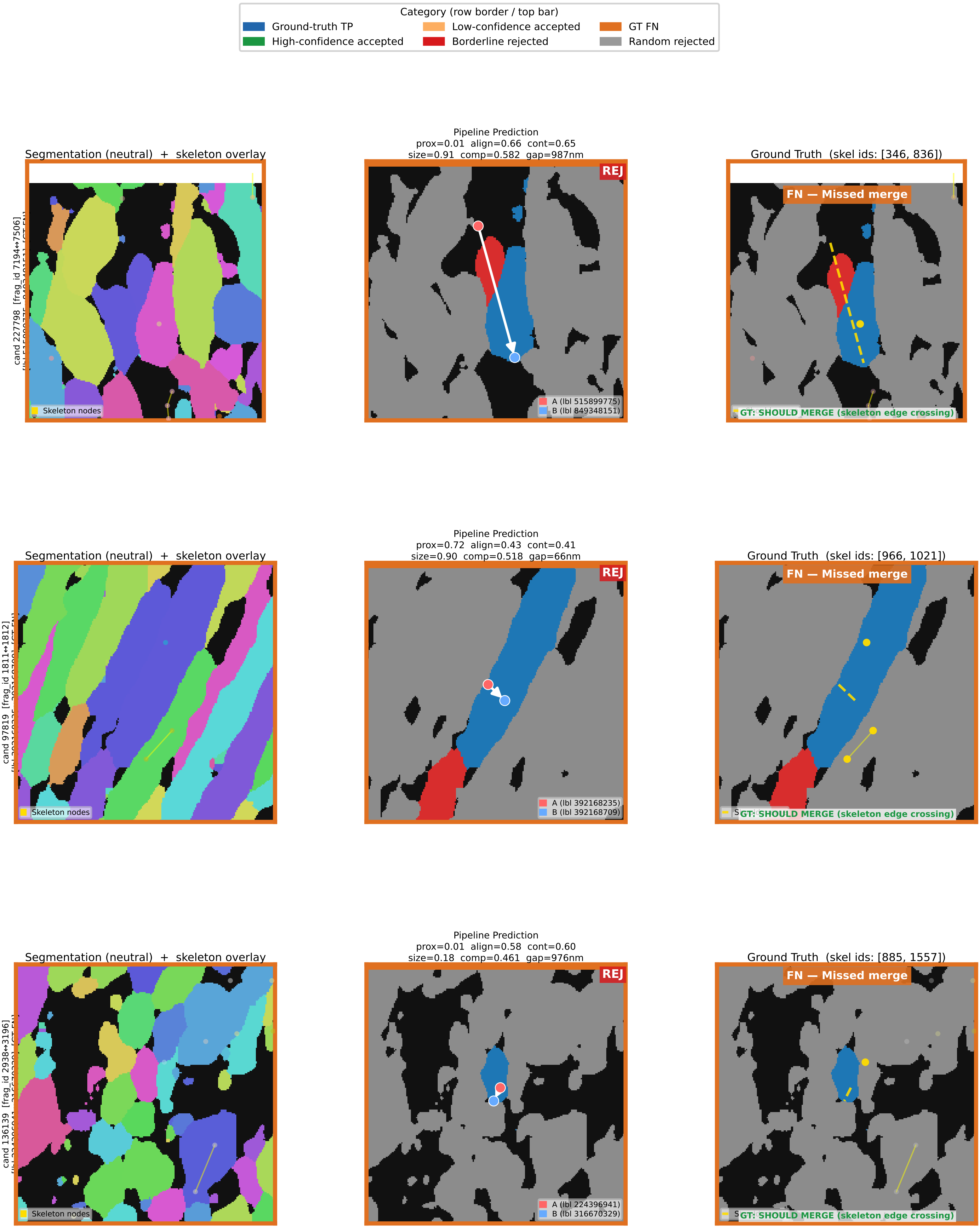

Category (row border / top bar)
Ground-truth TP
Low-confidence accepted
GT FN
High-confidence accepted
Borderline rejected
Random rejected
Segmentation (neutral) + skeleton overlay
cand 148138 [frag_id 3325+8423]
Skeleton nodes
Pipeline Prediction
prox=0.01 align=0.30 cont=0.32
size=0.00 comp=0.234 gap=859nm
ACC
A (lbl 427820217)
B (lbl 427820436)
Ground Truth
FP — False accept
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 127002 [frag_id 2522+2984]
Skeleton nodes
Pipeline Prediction
prox=0.02 align=0.30 cont=0.33
size=0.00 comp=0.238 gap=741nm
ACC
A (lbl 140509526)
B (lbl 285212803)
Ground Truth
FP — False accept
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 179232 [frag_id 5648+11589]
Skeleton nodes
Pipeline Prediction
prox=0.01 align=0.31 cont=0.32
size=0.00 comp=0.239 gap=1000nm
ACC
A (lbl 2993771573)
B (lbl 2993771686)
Ground Truth
FP — False accept
GT: NO MERGE



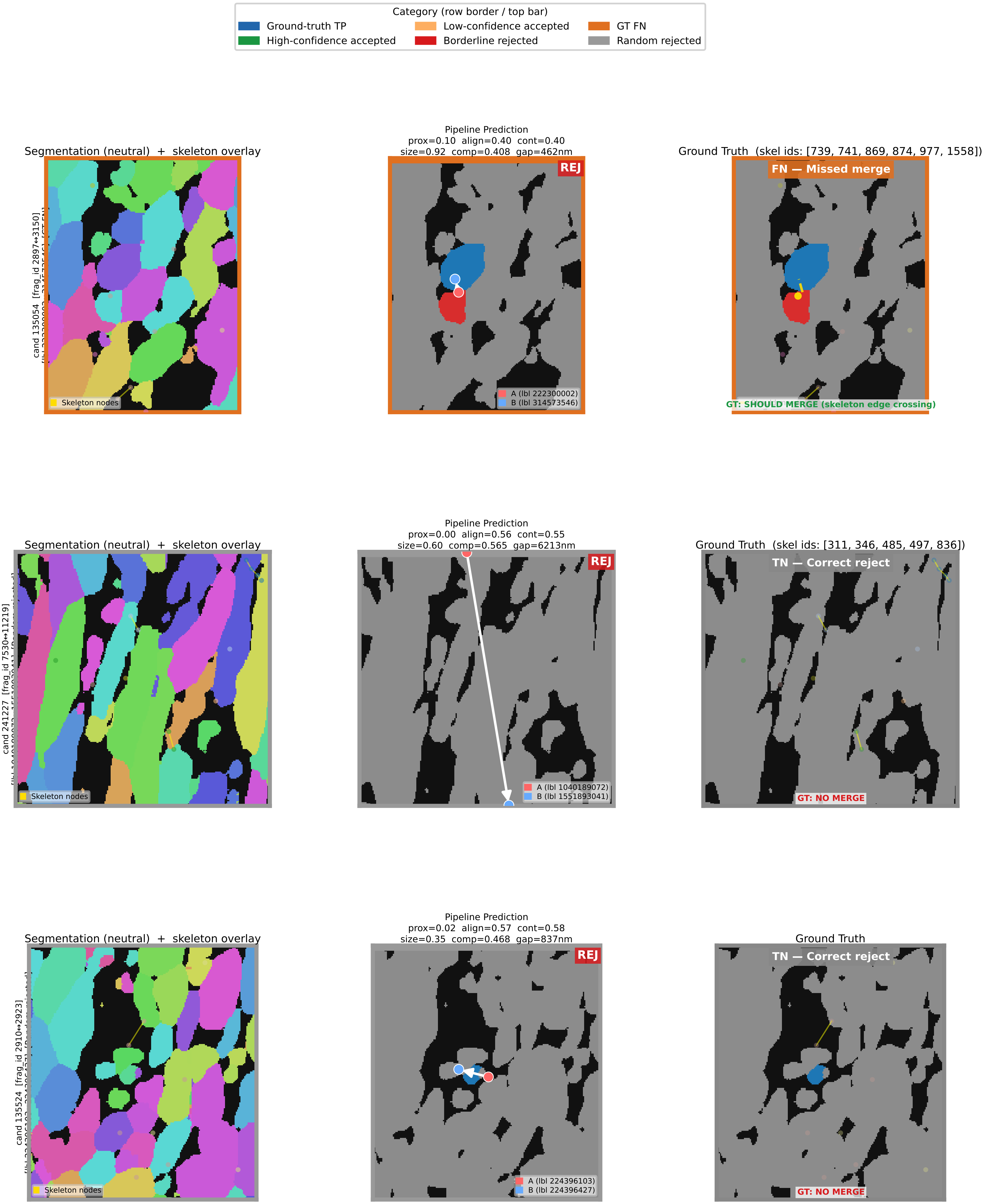
Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
cand 359672 [frag_id 13613↔13668]
Segmentation (neutral) + skeleton overlay
Skeleton nodes
Pipeline Prediction
prox=0.01 align=0.98 cont=0.92
size=0.92 comp=0.809 gap=913nm
REJ
A (lbl 1891746080)
B (lbl 2154676594)
Ground Truth
TN — Correct reject
GT: NO MERGE
cand 29373 [frag_id 696↔864]
Segmentation (neutral) + skeleton overlay
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.77 cont=0.69
size=0.99 comp=0.774 gap=1847nm
REJ
A (lbl 144703964)
B (lbl 203424939)
Ground Truth (skel ids: [417])
TN — Correct reject
GT: NO MERGE
cand 279843 [frag_id 8975↔13112]
Segmentation (neutral) + skeleton overlay
Skeleton nodes
Pipeline Prediction
prox=0.58 align=0.92 cont=0.83
size=0.25 comp=0.771 gap=109nm
REJ
A (lbl 704643841)
B (lbl 889194231)
Ground Truth
TN — Correct reject
GT: NO MERGE



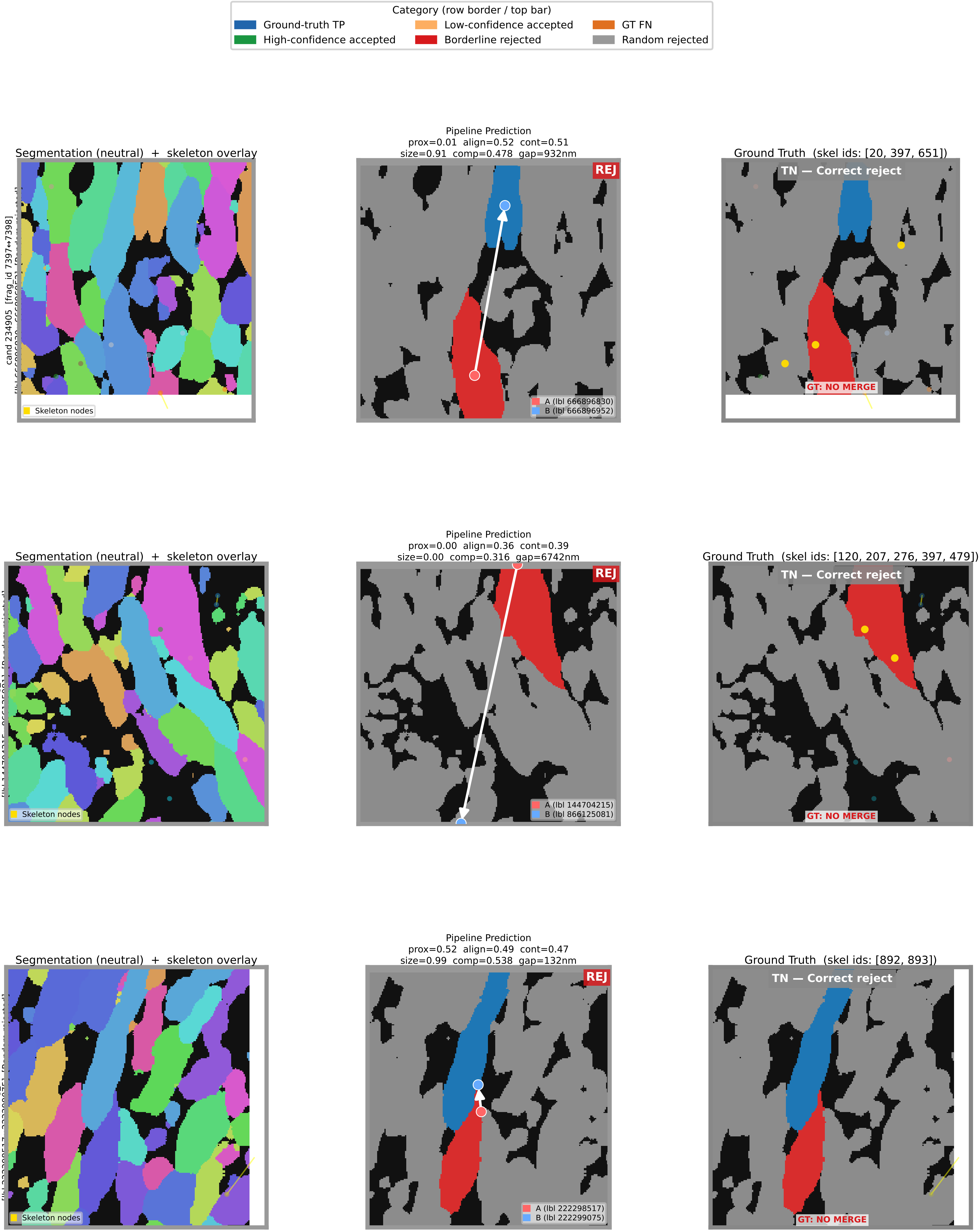
Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 29372 [frag_id 696↔859]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.74 cont=0.67
size=0.98 comp=0.745 gap=1441nm
REJ
A (lbl 144703964)
B (lbl 203424477)
Ground Truth (skel ids: [120, 207, 276, 479])
TN — Correct reject
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 221270 [frag_id 6949↔9374]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.74 cont=0.67
size=0.94 comp=0.741 gap=1941nm
REJ
A (lbl 385876625)
B (lbl 654311976)
Ground Truth (skel ids: [40, 41, 73, 78])
TN — Correct reject
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 113976 [frag_id 1929↔1931]
Skeleton nodes
Pipeline Prediction
prox=0.35 align=0.66 cont=0.65
size=0.97 comp=0.641 gap=209nm
REJ
A (lbl 1035994497)
B (lbl 1035994864)
Ground Truth (skel ids: [208, 249, 276, 575, 1004])
FN — Missed merge
GT: SHOULD MERGE (skeleton edge crossing)



Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 227798 [frag_id 7194↔7506]
Skeleton nodes
Pipeline Prediction
prox=0.01 align=0.66 cont=0.65
size=0.91 comp=0.582 gap=987nm
REJ
A (lbl 515899775)
B (lbl 849348151)
Ground Truth (skel ids: [346, 836])
FN — Missed merge
GT: SHOULD MERGE (skeleton edge crossing)
Segmentation (neutral) + skeleton overlay
cand 97819 [frag_id 1811↔1812]
Skeleton nodes
Pipeline Prediction
prox=0.72 align=0.43 cont=0.41
size=0.90 comp=0.518 gap=66nm
REJ
A (lbl 392168235)
B (lbl 392168709)
Ground Truth (skel ids: [966, 1021])
FN — Missed merge
GT: SHOULD MERGE (skeleton edge crossing)
Segmentation (neutral) + skeleton overlay
cand 136139 [frag_id 2938↔3196]
Skeleton nodes
Pipeline Prediction
prox=0.01 align=0.58 cont=0.60
size=0.18 comp=0.461 gap=976nm
REJ
A (lbl 224396941)
B (lbl 316670329)
Ground Truth (skel ids: [885, 1557])
FN — Missed merge
GT: SHOULD MERGE (skeleton edge crossing)



Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 135054 [frag_id 2897↔3150]
Skeleton nodes
Pipeline Prediction
prox=0.10 align=0.40 cont=0.40
size=0.92 comp=0.408 gap=462nm
REJ
A (lbl 222300002)
B (lbl 314573546)
Ground Truth (skel ids: [739, 741, 869, 874, 977, 1558])
FN — Missed merge
GT: SHOULD MERGE (skeleton edge crossing)
Segmentation (neutral) + skeleton overlay
cand 241227 [frag_id 7530↔11219]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.56 cont=0.55
size=0.60 comp=0.565 gap=6213nm
REJ
A (lbl 1040189072)
B (lbl 1551893041)
Ground Truth (skel ids: [311, 346, 485, 497, 836])
TN — Correct reject
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 135524 [frag_id 2910↔2923]
Skeleton nodes
Pipeline Prediction
prox=0.02 align=0.57 cont=0.58
size=0.35 comp=0.468 gap=837nm
REJ
A (lbl 224396103)
B (lbl 224396427)
Ground Truth
TN — Correct reject
GT: NO MERGE



Category (row border / top bar)
Ground-truth TP
High-confidence accepted
Low-confidence accepted
Borderline rejected
GT FN
Random rejected
Segmentation (neutral) + skeleton overlay
cand 234905 [frag_id 7397↔7398]
Skeleton nodes
Pipeline Prediction
prox=0.01 align=0.52 cont=0.51
size=0.91 comp=0.478 gap=932nm
REJ
A (lbl 666896830)
B (lbl 666896952)
Ground Truth (skel ids: [20, 397, 651])
TN — Correct reject
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 31271 [frag_id 706↔9055]
Skeleton nodes
Pipeline Prediction
prox=0.00 align=0.36 cont=0.39
size=0.00 comp=0.316 gap=6742nm
REJ
A (lbl 144704215)
B (lbl 866125081)
Ground Truth (skel ids: [120, 207, 276, 397, 479])
TN — Correct reject
GT: NO MERGE
Segmentation (neutral) + skeleton overlay
cand 133467 [frag_id 2854↔2867]
Skeleton nodes
Pipeline Prediction
prox=0.52 align=0.49 cont=0.47
size=0.99 comp=0.538 gap=132nm
REJ
A (lbl 222298517)
B (lbl 222299075)
Ground Truth (skel ids: [892, 893])
TN — Correct reject
GT: NO MERGE